Predicting Singapore HDB Rental Prices under Temporal Distribution Shift

Ox Alpha (Group 6)
Li Chengxuan (A0352428A), Mao Huanqing (A0257997Y)
Sun Zekun (A0351314R), Zhang Ziqing (A0223145R)
National University of Singapore

***Abstract*—**
Writing prompt: Write approximately 150–180 words after the final experiments are complete. State the practical rental-prediction problem, the dataset and temporal challenge, the analysis and preprocessing approach, the evaluated methods, and the selected final model. Give its verified evaluation results and the most important interpretation. Describe completed work in the past tense. Do not present progress-report scores as final results.

***Index Terms*—HDB rental prediction, temporal validation, feature engineering, regression, spatial auxiliary data**

I. Introduction

A. Motivation and Problem Context

Accommodation costs place a substantial financial burden on households in Singapore. As the course overview emphasises, renting can be a practical necessity for people who are unable or ineligible to purchase housing, including singles, young couples waiting for a new flat, and foreign workers [1]. Renters need to judge what their budget can secure and compare asking rents across properties; landlords and real estate agents need realistic pricing references. These needs motivate our study of Housing & Development Board (HDB) flats. Given a flat's characteristics and approval month, we predict its monthly rent from historical observations and assess the reliability of predictions for later periods.

This task presents three challenges. First, rental levels change over time: in the supplied training data, mean rent rises from SGD 2,121 in 2021 to SGD 3,036 in 2023, followed by much smaller changes in 2024 and early 2025. These descriptive averages also reflect the mix of properties observed, so they do not isolate a causal market effect. Nevertheless, the pattern makes continued extrapolation of an earlier growth trend questionable. Second, location and property characteristics are interdependent: town, street, flat type and floor area may encode overlapping information, while amenity features may contribute differently across models. Third, repeated feature vectors and inconsistent derived attributes require careful treatment. Without identifiers for individual flats or contracts, matching rows cannot be assumed to represent duplicate transactions.

Useful rental estimates must remain credible when the prediction period extends beyond the observed market conditions. An underestimated rent can mislead a tenant's budget, while an overestimated rent can distort property comparisons or pricing expectations. Moreover, low average error can conceal poor performance for uncommon flat types or less represented locations. We therefore consider temporal calibration and subgroup errors alongside overall accuracy. This framing treats the model as a reference based on observable attributes, rather than a complete valuation of an individual flat's condition or contractual terms.

B. Research Questions and Contributions

We address three questions: (1) how do temporal, spatial and property patterns inform preprocessing and rent prediction; (2) what additional predictive value do contextual and historical-rent features provide beyond the core attributes; and (3) which evaluated model performs most reliably on later observations, and where do its errors remain largest?

Our work connects exploratory analysis to explicit preprocessing decisions, compares alternative treatments of repeated records, and evaluates Ridge regression, CatBoost and a multilayer perceptron under chronological validation. Feature comparisons retain the same individual validation records and original rents. Data-dependent transformations and rental-history summaries are constructed from the training portion of each experiment. The selected model is assessed through later-period errors and a longer forecasting-horizon diagnostic, rather than through an aggregate score alone. These contributions concern the application and evaluation of existing methods; their value lies in identifying which choices support reliable prediction for this dataset.

II. Data and Problem Formulation

A. Prediction Task and Core Data

The CS5228 competition supplies labelled training records and an unlabelled test set [1]. We learn a regression function $\hat{y} = f(x)$, where y is `MONTHLY_RENT` in Singapore dollars (SGD). The ten supplied predictors describe the approval month, town, block, street, flat type, flat model, floor area, furnishing status, lease commencement year and upfront fee. All 150,000 training rows are retained; preprocessing is detailed in Section III.

As Table I shows, the split is predominantly temporal: only 380 test records share March 2025 with the training period; all others are later. The maximum test horizon is 16 months beyond the last training month. We assess prediction error primarily with root mean squared error (RMSE) in SGD; the evaluation protocol is specified in Section V.

TABLE I
OVERVIEW OF THE SUPPLIED RENTAL DATA.

Property	Training data	Test data
Raw records	150,000	50,000
Rental months	Jan. 2021–Mar. 2025	Mar. 2025–Jul. 2026
Supplied predictors	10	10
Target labels	Available	Unavailable
Final modelling records	150,000	50,000

B. Auxiliary Data Sources

Six supplied auxiliary datasets cover HDB block metadata, Mass Rapid Transit (MRT) stations, schools, shopping malls, Certificate of Entitlement (COE) [2] prices, and stock prices for major Singapore companies [3]. Addresses and coordinates support building attributes and amenity features. The selected feature set combines core, HDB/MRT and training-only rental-history features. School and mall features were evaluated but excluded. COE and stock features were excluded because future observations are unavailable at the prediction cutoff. Historical snapshot limitations are discussed in Section III.

III. EXPLORATORY DATA ANALYSIS AND PREPROCESSING

A. Target Distribution and Data Quality

The 150,000 raw training records contain no missing cells. Monthly rent has a mean of SGD 2,719, a median of SGD 2,700, and a range of SGD 300 to SGD 7,600. Its skewness is 0.39. The interquartile-range rule flags 752 records (0.50%) as extreme, but the supplied attributes cannot establish that these are recording errors. We retained them and used all original rents for evaluation. Ridge and CatBoost learned from raw SGD targets; the MLP standardised targets within each training fold and converted predictions back to SGD. No logarithmic target transformation was applied.

Two predictors have no variation: `FURNISHED` is always `yes`, and `FEE` is always zero. We removed both. Raw completeness does not eliminate the need for missing-value handling: masking invalid ages and cleaning sentinels in the auxiliary tables introduce missing values later in the pipeline.

B. Temporal Patterns and Distribution Shift

Fig. 1 shows rapid rent growth followed by a flatter period. Mean rent rises from SGD 2,121 in 2021 to SGD 3,036 in 2023, then changes less in 2024 (SGD 3,110) and January to March 2025 (SGD 3,142). The last value covers only three months. The average of the first six monthly means is SGD 2,079, compared with SGD 3,137 for the final six. These changes motivate a continuous month index and calendar features, but also make extrapolating an earlier growth rate a potential source of error. We therefore trained on earlier months and evaluated on later months, as specified in Section V.

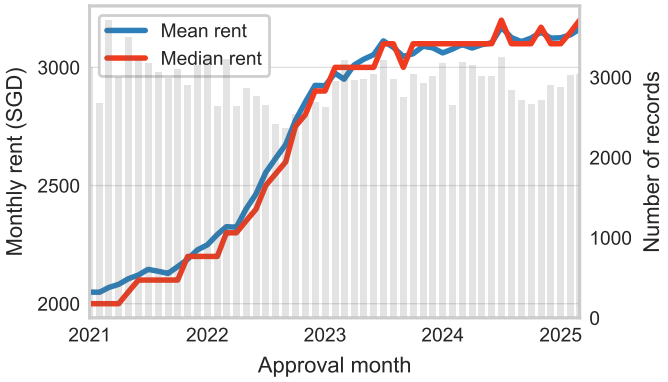


Fig. 1. Monthly mean and median rent across all 150,000 raw training records. Faint bars show record counts on the right axis. Early 2025 covers January to March only.

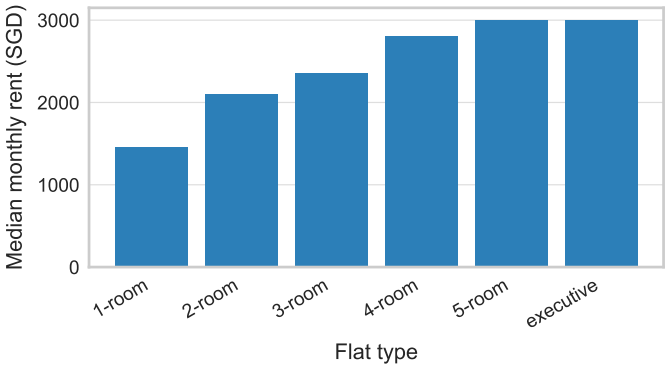


Fig. 2. Median rent by normalised flat type across the raw training records. The comparison does not adjust for month, location, or floor area.

C. Spatial and Property Patterns

Rent varies across both property types and towns. After normalising flat-type labels, median rent increases from SGD 2,100 for two-room flats to SGD 3,000 for five-room and executive flats (Fig. 2). Floor area has a positive raw correlation with rent (Pearson $r = 0.33$), although it is closely related to flat type. We retained both descriptors. Town comparisons also differ with size: Central has a median rent per square metre of SGD 41.10, compared with SGD 23.44 in Pasir Ris, whose mean floor area is larger. These comparisons pool different months and property mixes; they support location features without identifying causal location premiums. Rent per square metre was used only for EDA because it contains the target.

D. Core Cleaning and Feature Construction

We applied Unicode NFKC normalisation, lowercasing, and whitespace cleanup to categorical fields, removed spaces within block identifiers, and merged equivalent labels such as `4 room` and `4-room`. Flat-type cardinality fell from eleven to six, and street cardinality from 1,146 to 576. Space-separated room labels occur mainly in 2021, so leaving the spellings separate would mix property categories with formatting changes over time.

TABLE II
EDA FINDINGS AND THE CORRESPONDING PREPROCESSING DECISIONS.

Finding	Decision and justification
Constant fields	Remove fields without predictive variation.
Inconsistent category labels	Normalise case, whitespace and equivalent room labels.
Changing rent trend	Encode approval time and validate on later months.
Invalid ages and sentinels	Mask source values, attempt training-only address recovery, and retain quality flags.
Identical observations	Retain all rows; exact removal gave negligible benefit in controlled validation.
Extreme targets	Retain rents flagged by IQR without evidence of error.

Approval month was parsed into year, month, quarter, a month index, and sine/cosine seasonal terms. We added town/street and town/block/street keys for address resolution, an ordered flat-type code, lease age, its invalidity flag, and $\log(1 + \text{floor area})$ alongside raw area. These transformations give 19 core predictors. Original row IDs and parsed dates remain alignment and validation fields, not model inputs.

Table II summarises the cleaning decisions. We checked the uncertain choices through the preprocessing audit below rather than assuming that fewer records or more elaborate imputation would improve prediction.

E. Preprocessing Audit and Policy Selection

The audit in Notebook 06 compared three record policies and two missing-value treatments using Ridge with $\alpha = 10$ and the same 35 core/HDB/MRT features. Selection used July to December 2023 and January to June 2024, training on earlier months in each case. Record policies changed the training data only. The two validation blocks retained all 17,900 and 18,179 original observations, respectively, and their individual rent labels. Table III reports the unweighted mean RMSE across these two windows.

1) Repeated observations: The raw data contain 425 exact matches beyond the first occurrence; normalisation raises this count to 641. Unit numbers are unavailable, so identical attributes and rent cannot establish that a transaction was recorded twice. Multiple units in the same block can share the supplied description. The audit also found 21,974 repeated feature rows when rent was excluded from the grouping key. The largest group contains ten observations, and 17,900 feature groups have more than one rent value. These counts distinguish identical complete records from repeated descriptions with different targets.

We tested retaining all rows, removing exact matches including rent, and collapsing each identical feature vector to one row with mean rent. The last policy reduces a group's contribution to training because no group-size weight is retained. Its mean RMSE was about 603.96 SGD, compared with about 600.80 SGD for the other policies. We did not average validation labels under any policy.

TABLE III
NOTEBOOK 06 PREPROCESSING AUDIT. MEAN RMSE IN SGD OVER THE TWO EARLIER SELECTION WINDOWS; VALIDATION OBSERVATIONS ARE IDENTICAL FOR ALL ROWS.

Training record policy	Median only	Mask/recover
Retain all	600.803	600.808
Remove exact matches	600.798	600.792
Collapse to mean rent	603.965	603.962

2) Invalid values and address recovery: Lease age is approval year minus lease commencement year. Ages below zero or above 100 years are invalid under our rule. One training record is affected: a November 2021 approval at Bedok block 153B, Bedok South Road, has lease commencement year 2022. The masking/recovery variant removes this source value as well as its derived age. HDB maximum floors and completion years at or below zero are also masked, and completion years are checked for an implied age outside the same range. Each HDB source field is invalid or missing for 159 training records across eleven addresses. None of these audited numeric fields is invalid or missing in the supplied test rows. These checks identify values the model should not use directly; they do not determine corrected values.

Recovery uses the full normalised town/block/street key. A source attribute can be recovered only when all valid training values at that address agree. Conflicting values and unseen addresses retain missing values. We fit the lookup within each training fold, recheck recovered years against the approval year, recompute ages, and keep the original invalidity flags. Remaining numeric gaps use training-fold medians for Ridge and MLP, or CatBoost's native missing-value handling.

The recovery audit filled no source values in either selection window. In the first validation block, nine records lacked maximum-floor values and nine lacked completion years; all remained missing. The second block had no missing audited source attributes. Since HDB attributes come from a shared block lookup, another transaction at the same address need not supply an independent valid value. The result gives no evidence of a predictive gain from recovering attributes in these windows.

The selection rule preferred retaining records when its mean RMSE was within 1 SGD of the best policy, and preferred masking/recovery when within 0.1 SGD of the best missing-value treatment. Retention with masking/recovery scored 600.808 SGD, only 0.016 SGD above exact removal with the same treatment. Its score was also 0.005 SGD above retention with median-only handling. We selected it to preserve potentially valid transactions and mask invalid source years, rather than claiming an accuracy improvement from imputation. These tolerances are practical selection rules, not significance thresholds. The final pipeline therefore retains all 150,000 training records and every one of the 50,000 test IDs.

F. Auxiliary Construction and Record Integrity

The 9,464 HDB block records have unique normalised town, block, and street keys, with 100% address coverage in train and test. A many-to-one join adds coordinates, maximum floor, completion year, postal sector, subzone, planning area, region, and derived age/quality fields. The eleven blocks with invalid floor/year sentinels account for the 159 affected training records; 285 block postal codes marked `NIL` become missing categories.

Rail data provide current status but no opening dates. We used a fixed proxy of older line prefixes, excluded eight specified later/future codes, and deduplicated station names and coordinates, leaving 122 locations. The proxy excludes TEL and LRT and does not track openings during the rental period. Haversine distances yield the nearest-station distance and counts within 500 metres and one kilometre. School and mall distances/counts were built similarly and tested as separate feature groups. They were excluded from the selected configuration; their snapshots also lack opening histories.

We inspected the supplied Certificate of Entitlement (COE) and stock-price tables, linked in the project description to COE background [2] and a Singapore company ranking [3]. Their cleaned observations were aggregated by calendar month and shifted by one month. Neither source entered the final forecasting model: a lag does not establish availability of market observations throughout the future prediction horizon. We collected no additional data.

The final feature set combines the 35 core/HDB/MRT inputs with four rental-history fields: past mean rent and count for town/flat-type pairs and for properties. For training rows, histories exclude the current month and all later months. Validation and test histories use only rents known in their training data. The final contract therefore has 39 predictors. Joins check key uniqueness, row counts, and matching IDs/dates so that adding features cannot silently change the evaluation population.

IV. Data Mining Methods and Model Development

A. Reference Predictor and Ridge Baseline

Writing prompt: Describe the historical town–flat-type median reference and its fallback rules. Explain why Ridge was selected as a fast baseline for sparse, correlated predictors. State its feature set and target representation. Describe how it was applied rather than reproducing textbook derivations.

B. Candidate Models

Writing prompt: Describe the alternative models actually evaluated after the progress report. Possible candidates include boosting models, but do not list a method as an experiment unless it was run. For each model, explain its suitability, categorical handling, target representation and principal configuration. Add verified references for advanced methods. Remove unused candidates.

C. Completed Model Improvements

Writing prompt: Use this subsection for the improvements completed before submission. For each change, connect an observed weakness to a specific modification and its controlled evaluation. Report unsuccessful changes when they explain why a component was excluded. Do not write this subsection as a plan.

1) Temporal Representation and Calibration: Writing prompt: Describe any completed changes to linear time trends, piecewise or capped trends, training windows, recency weights or calibration. Explain how cutoffs and other choices were selected using historical validation. State which approach the final model retained. Delete this heading if no such experiment was completed.

2) Feature Interactions and Auxiliary Information: Writing prompt: Describe completed tests of time–location or property interactions, alternative categorical representations, and additional auxiliary feature groups. Identify retained and rejected features. Distinguish a plausible explanation from an effect demonstrated by ablation.

3) Tuning and Model Combination: Writing prompt: Describe completed tuning, early stopping or ensemble experiments. If an ensemble was used, specify its component models, training protocol and how its weights were selected without using evaluation labels. Remove this heading when the final method is a single model with no relevant experiment.

D. Final Model Specification

Writing prompt: Give the selected final model or ensemble, exact feature groups, target transformation, hyperparameters and training procedure. Explain the selection criterion. This subsection must describe the implemented final system, not merely the highest-scoring intermediate run. State the final refitting procedure and any post-processing used for test predictions.

V. Experimental Setup

A. Chronological Validation

Writing prompt: Explain why evaluation must train on earlier rentals and predict later rentals. Table IV shows the existing expanding windows. Replace or supplement them if the final protocol changes. State the prediction horizon and any gap. Explain how final performance was evaluated separately from tuning, or explicitly disclose reuse of the tuning folds.

Writing prompt: The current validation blocks span 6, 6 and 9 months, while the maximum test prediction distance is 16 months after the last training month. Describe any completed longer-horizon evaluation, or acknowledge the horizon limitation. A fold already used to choose models is not an untouched holdout.

TABLE IV
EXISTING CHRONOLOGICAL VALIDATION WINDOWS. UPDATE TO MATCH THE FINAL EVALUATION PROTOCOL.

Fold	Training through	Validation period
1	Jun. 2023	Jul.–Dec. 2023
2	Dec. 2023	Jan.–Jun. 2024
3	Jun. 2024	Jul. 2024–Mar. 2025

B. Training-Fold Preprocessing and Leakage Controls

Writing prompt: Describe how numerical imputation, scaling, categorical vocabularies and infrequent-category grouping were fitted

Figure placeholder

Per-fold RMSE for the baseline and final model.

Fig. 3. Performance across chronological validation periods. Replace the placeholder with results from the final experiment runs.

Figure placeholder

Monthly actual and predicted mean rent for the baseline and final model on the same latest evaluation period.

Fig. 4. Temporal calibration of the baseline and final model. Report mean bias and the validation period in the completed caption.

inside each historical training fold. Explain any model-specific exceptions. Exclude target-derived inputs. State how unknown categories and auxiliary missingness were handled.

C. Hyperparameter Selection and Reproducibility

Writing prompt: Report the actual search strategy, search space, validation criterion, random seeds, software versions and relevant resource constraints. The existing Ridge search uses alpha in {0.1, 1, 10, 100}; update this if expanded. Summarise decisive settings in the main text and put the full grid in the appendix. Every result must be traceable to a saved experiment configuration.

D. Metrics and Controlled Comparisons

Writing prompt: Use RMSE as the primary metric and MAE and R-squared as supporting diagnostics. Define whether results are unweighted means of fold metrics or pooled validation metrics. Use identical record IDs and original labels for feature comparisons. State the tuning protocol for model comparisons. If uncertainty is reported, specify how temporal dependence was handled.

VI. Final Results and Discussion

A. Overall Model Comparison

Writing prompt: Fill Table V with the final verified results from one consistent evaluation protocol. Compare the reference, baseline, improved model and selected final configuration. Explain why the final model was selected, including performance consistency and relevant trade-offs. All values below remain TBD until the final runs are frozen.

B. Performance across Time Periods

Writing prompt: Report per-fold results and explain differences between market periods. A lower average alone is insufficient to establish temporal robustness. Figure 3 should compare the principal methods across the same validation windows.

C. Ablation Studies

Writing prompt: Quantify the contribution of completed improvements by adding or removing one component while keeping the evaluation population fixed. Select the most informative feature, temporal and modelling ablations. Table VI is a template, not a required list of experiments.

Writing prompt: The progress report's approximately 0.1% HDB/MRT gain is an intermediate Ridge finding. Re-evaluate its contribution under the final model and describe tiny differences proportionately. Do not call a difference statistically significant without an appropriate uncertainty analysis.

D. Error Analysis and Model Interpretation

Writing prompt: Analyse the final model's errors by month, flat type, town and rent range. Report subgroup sample sizes, RMSE or MAE, and signed bias. Define residual as actual minus predicted. Show whether the final model still overpredicts the late-period plateau and compare with the Ridge baseline. Explain poor cases with observed evidence, while distinguishing unobserved causes and hypotheses. Treat correlated coefficients or feature attributions as predictive descriptions, not causal effects.

E. Final Test Predictions and Submission

Writing prompt: Identify the final refit, submission configuration and 50,000 predictions aligned to the sample-submission IDs. Report an actual competition score only if available, specifying public/private split and the selected submission. Do not infer test accuracy from prediction ranges or compare a competition score directly with a different internal validation population.

F. Limitations of the Final Findings

Writing prompt: Discuss limitations supported by the completed analysis: missing unit-level attributes, small subgroups, duplicate-record uncertainty, historical snapshot accuracy, correlated features and evaluation-horizon coverage. Explain how these constrain the claims about the selected model. Keep this as an assessment of the final evidence, without a future-outlook or planned-experiments section.

VII. Conclusion

Writing prompt: Answer the research questions using the completed final results. Name the selected model, quantify its performance relative to the baseline, state which improvements and auxiliary features proved useful, and summarise the most important error or limitation. End with the practical interpretation of the findings. Do not end with a list of work the team intends to do.

TABLE V
FINAL COMPARISON UNDER THE COMMON EVALUATION PROTOCOL. REPLACE GENERIC MODEL NAMES AND REMOVE ROWS FOR EXPERIMENTS NOT COMPLETED.

Method	Feature set	RMSE (SGD)	MAE (SGD)	R^2	RMSE reduction (%)
Historical median	Town and flat type	*TBD*	*TBD*	*TBD*	—
Ridge baseline	Core	*TBD*	*TBD*	*TBD*	*TBD*
Ridge baseline	Core + HDB/MRT	*TBD*	*TBD*	*TBD*	*TBD*
Improved model	Specify evaluated inputs	*TBD*	*TBD*	*TBD*	*TBD*
Final selected model	Specify final inputs	*TBD*	*TBD*	*TBD*	*TBD*

Specify the score aggregation and evaluation population. Define percentage reduction relative to the Core Ridge baseline. If the improved and final models are identical, combine those rows. Use a clearly separate column for any external competition score.

TABLE VI
CONTROLLED ABLATION OF THE FINAL MODEL.

Configuration	RMSE (SGD)	ΔRMSE
Full final configuration	*TBD*	0.00
Without auxiliary spatial features	*TBD*	*TBD*
Without the selected temporal change	*TBD*	*TBD*
Without a tested interaction/component	*TBD*	*TBD*

Define ΔRMSE as the ablated configuration's score minus the full configuration's score. Remove rows for components not tested.

REFERENCES

[1] National University of Singapore, "CS5228 course project," Kaggle competition and supplied rental and auxiliary datasets. [Online]. Available: https://www.kaggle.com/competitions/cs-5228-2610-project

[2] Wikipedia contributors, "Certificate of Entitlement," *Wikipedia*. [Online]. Available: https://en.wikipedia.org/wiki/Certificate_of_Entitlement. Accessed: Oct. 10, 2026.

[3] CompaniesMarketCap, "Largest companies in Singapore by market capitalization." [Online]. Available: https://companiesmarketcap.com/sgd/singapore/largest-companies-in-singapore-by-market-cap/. Accessed: Oct. 10, 2026.

[4] *Placeholder: verified reference for Ridge regression, if cited.*

[5] *Placeholder: original paper or authoritative reference for the advanced method actually used in the final model.*

Appendix A
Team Workload Breakdown

Writing prompt: Record actual completed contributions. The names, IDs and team name come from the supplied progress slides. Confirm them before submission. Only Sun Zekun's core-preprocessing responsibility is prefilled from the conversation; complete all other entries from the team's actual workload.

Appendix B
Supplementary Details

Writing prompt: Use sparingly. Include only the full hyperparameter search/configuration table, a compact feature dictionary, or supplementary EDA/error tables that help reproduce or elaborate the main results. Do not move the final model definition, validation design, main results or essential interpretation out of the main text. The paper must remain comprehensible without this appendix.

TABLE VII
TEAM WORKLOAD AND REPORT CONTRIBUTIONS.

Member	Student ID	Main project responsibilities	Report contributions
Li Chengxuan	A0352428A	*TBD*	*TBD*
Mao Huanqing	A0257997Y	*TBD*	*TBD*
Sun Zekun	A0351314R	Core data preprocessing; complete other actual responsibilities	Core preprocessing; complete actual writing/review tasks
Zhang Ziqing	A0223145R	*TBD*	*TBD*